

INTEGRATED PRACTICE QUIZ:

Pharmacology of Aminoglycosides, Protein Synthesis Inhibitors, Folate Synthesis Inhibitors, and Fluoroquinolones

This quiz has lots of questions because it covers four major antibiotic classes and various frequently-occurring infections in various real-life contexts.

HINT: Identify the clues in the vignette that you need to answer the question. Don't be distracted by the information that is not needed to answer the question.

- **Suggestion for solving the puzzles:** Circle the clues and think about each option – reason how each option fits the clues and how it doesn't fit the clues.
It is just as important to understand why the correct option(s) is correct and the wrong options are incorrect.
Then, when the answers are posted, look at the explanations. At a later time (spacing practice), try it again without looking at the explanations.

Tips for successful learning:

The *purpose* of integrating four classes of antibiotics into one practice quiz is to *help you improve your knowledge and understanding of new material* by mixing up problem types (“interleaving”), rather than focusing on one type at a time before moving on to study another type.

Self-quizzing, spaced retrieval practice, interleaving, and other study practices that solidly embed knowledge help learners avoid the “illusion of knowing”.

When will the answers be posted?

The answers will be posted on the Friday before Thanksgiving break.

Meanwhile, all answers can be found by doing a search of the relevant Notes handouts.

Abbreviations:

Sig = signetur translated as “let it be labeled”

po = per os (by mouth); IM = intramuscular; IV = intravenously; IT = intrathecally; IO = intraosseously

Q12h= every 12 hours | Q8h= every 8 hours | TID= three times daily | BID= twice daily | QD= once daily

Learning Tip: Retrieval Practice means self-quizzing (before looking at the answers)

Self-quizzing by focusing on the main ideas and the meanings helps you to focus on the central principles and provides **a reliable measure** of what you have learned and what you have not yet mastered.

The quiz begins on the next page.

1. Which one of the following is a common feature among all the protein synthesis inhibitors?

- A. Action at the bacterial ribosome
- B. Bactericidal activity
- C. Low toxicity to host cells
- D. PK-PD profiles
- E. Renal elimination
- F. Spectrums of activity

2. What is the mechanism by which aminoglycosides gain access to bacterial cell cytoplasm?

- A. Active transport by oxygen-dependent process
- B. Active transport by iron-transport systems
- C. Endocytosis
- D. Passive diffusion through cytoplasmic lipid bilayer
- E. Passive diffusion via a solute carrier

3. Why are aminoglycosides effective in the treatment of susceptible infection with extended-interval once daily dosing even though the elimination half-life in adults with normal renal function is only about 2-3 hours?

Explain the actions and effects using the PK-PD model.

4. Which organisms would be intrinsically resistant to the aminoglycosides and by what mechanisms?

Short answer

5. A 10-year-old boy presents to the pediatrician with complaints of headache, fatigue, malaise, and muscle and joint aches for the past 5 days, and “a big bug bite” on his right leg. He and his friend were exploring in the woods outside of their Long Island town the weekend before last (10 days ago). NKDA.

Fever is absent and vital signs are normal. A single circular red area that clears in the middle, forming a bullseye pattern (erythema migrans), at the site of the tick bite is noted on physical exam. Lymphadenopathy is noted in the regional lymph nodes.

The patient is given a clinical diagnosis of Lyme disease. The patient states that he can swallow tablets. An e-prescription is submitted for a ten-day course of antibiotic therapy with an oral broad-spectrum drug effective in the treatment of many zoonotic diseases.



UpToDate: Erythema migrans Graphic 81270 Version 1.0 accessed from Lyme disease graphics.

The patient and his mother are advised that the drug can cause stomach upset and can be taken with food a liquid and the patient should not lie down for at least 30 minutes after taking the medicine to prevent irritation of “your food pipe, which is call the esophagus”. They are also cautioned about wearing long sleeves, long pants, a hat, and sunblock while in the sun as the drug may cause a bad sunburn-like reaction (photosensitivity).

i) Which one of the following options best represents the mechanism of action of the selected antibiotic?

- A. Prevents access of aminoacyl tRNA to the acceptor (A) site on the mRNA-ribosome complex
- B. Prevents assembly of the functional ribosome and causes misreading of the genetic code
- C. Prevents formation of the ribosomal-fMet-tRNA complex that initiates protein synthesis
- D. Prevents the interaction between peptidyl transferase and its amino-acyl tRNA
- E. Prevents the translocation of a newly synthesized peptidyl tRNA molecule from the acceptor site (A) to the peptidyl (P) donor site

Hint: Think about the cause of the infection and the group of drugs that are very effective against this classification of diseases. (Hint: key word: zoonotic)

ii) Where is the antibacterial binding site of this drug?

iii) What is the organism that causes Lyme disease?

iv) What treatment would be recommended if the child is 6 years old?

6. Patients are cautioned to avoid di- and trivalent cation-containing foods and supplements when taking an oral drug from the tetracycline class.

Why?

7. KC Jr. is 38-day-old uncircumcised male infant brought to the pediatrician due to fever of 10 hours, irritability, and poor feeding. He was delivered vaginally at 36 weeks gestational age (premature) and has been thriving since being discharge home with his parents. Temperature (rectal) is 100.8°F. All other physical findings are within normal limits. NKDA. A dipstick urinalysis is positive for leukocyte esterase and nitrite consistent with an *E. coli* infection. A urine sample is sent to the microbiology laboratory for culture and sensitivity testing. A sulfonamide antibiotic would likely be effective but should be avoided in this patient who is less than 2 months old.

What is the potential adverse effect that could occur with sulfonamide use in this patient?

- A. Organ damage due to accumulation of toxic drug metabolites
- B. Tooth discoloration due to chelation with calcium in tooth enamel
- C. Elevated plasma levels of free unconjugated bilirubin due to displacement from albumin binding sites
- D. Hemolytic anemia due to glucose-6-phosphate dehydrogenase deficiency
- E. Renal tubule irritation due to crystalluria

8. KC Jr. is evaluated for outpatient treatment. *E. coli* resistance to ampicillin is prevalent in the region. An e-prescription is submitted for the oral suspension of a narrow-spectrum beta-lactam from a different subclass. The drug has good bioavailability and good activity against the infective pathogen: Sig: 75 mg/kg po TID x10 days.

What is the drug?

Big hints: 1) narrow spectrum, 2) the mnemonic for the drug's Gram-negative activity is PECK.

FYI: You do not need to know dosing for now. You might find it helpful to get a sense of how drugs are prescribed.

(UpToDate Algorithm: Choice of empiric antibiotic to treat urinary tract infection in children 1 month to 2 years.)

9. A 17-year-old female presents to the dermatology clinic with moderate inflammatory acne affecting her face and upper back. She has tried over-the-counter benzoyl peroxide and topical retinoids with limited success. The dermatologist prescribes an oral antibiotic in the tetracyclines class as part of her treatment plan.

After six weeks of therapy, her acne has improved significantly. However, she reports experiencing occasional dizziness and a dark discoloration on her gums.

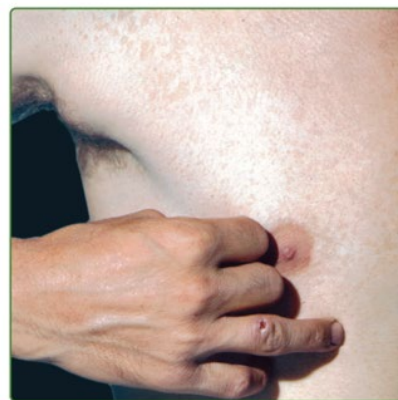
What is the name of the drug?

- A. Demeclocycline
- B. Doxycycline
- C. Minocycline
- D. Tetracycline
- E. Tigecycline

10. A 16-year-old male presents to the ED with sudden onset of fever, chills, anorexia, and malaise. He also had a headache, chest soreness, abdominal pain and vomiting earlier today but these symptoms have abated. He went rabbit hunting 3 days ago with his dad. The patient was in charge of cleaning and preparing the rabbits for cooking.

The patient's temperature is 104°F; a chancre-like ulcer with raised margins is noted on the dorsum of the fourth digit of his right hand and regional axillary lymphadenopathy is noted (see photo).

Blood samples are drawn for serology for suspicion of *Francisella tularensis* (a non-enteric aerobic Gram-negative coccobacillus).



Source: S. Kang, M. Amagai, A.L. Bruckner, A.H. Enk, D.J. Margolis, A.J. McMichael, J.S. Orringer: Fitzpatrick's Dermatology, Ninth Edition Copyright © McGraw-Hill Education. All rights reserved.

Tularemia. A chancre-like ulcer with raised margins on the dorsum of the fourth digit with accompanying axillary lymphadenopathy. The rash on the chest is unrelated and is pityriasis versicolor.



Citation: Chapter 156 Miscellaneous Bacterial Infections with Cutaneous Manifestations. Kang S, Amagai M, Bruckner AL, Enk AH, Margolis DJ, McMichael AJ, Orringer JS. Fitzpatrick's Dermatology. 9e. 2019. Available at: <https://accessmedicine.mhmedical.com/content.aspx?bookid=2578§ionid=270431229>. Accessed November 06, 2020. Copyright © 2020 McGraw-Hill Education. All rights reserved.

The patient is told that “rabbit fever” can be serious and the treatment associated with excellent rates of cure and minimal relapses by shots into the muscle of the buttocks or thigh every 12 hours for 10 days. A nurse will come to the home twice daily to administer the intramuscular injections, take blood samples for monitoring kidney function and drug levels, and monitor side effects, which include injection site reactions and nerve toxicities. The patient is advised to report ringing in his ears (tinnitus), dizziness (vertigo), tingling or numbness in his extremities, and diminished vision or a blind spot (scotoma).

Which of the following classes best fits the description of the administered drug?

- A. Aminoglycosides
- B. Beta-lactams
- C. Chloramphenicol
- D. Cyclic lipopeptides
- E. Folate synthesis inhibitors
- F. Fluoroquinolones
- G. Glycopeptides
- H. Lincosamides
- I. Oxazolidones
- J. Macrolides
- K. Tetracyclines

11. A normally healthy 34-year-old woman at 14 weeks gestation presents to the emergency department in Jonesboro, AK with fever, headache, acute abdominal pain, nausea, vomiting, cough, myalgia, malaise, and conjunctivitis. The symptoms came on suddenly last night, about 18 hours ago. She and her husband returned from the Ozark Folk Festival 8 days ago; they stayed at a campground near the event. She does not recall noticing a tick bite.

Blood samples are obtained for indirect immunofluorescence assay testing for *Rickettsia rickettsii*.

Empiric IV antibiotic therapy is initiated based on clinical suspicion of Rocky Mountain spotted fever. The selected drug is metabolized in the liver by glucuronidation and excreted in urine as glucuronide metabolites.

i) What antibiotic was administered to this patient?

ii) Because of potentially serious drug-specific toxicities of the conventional drug, monitoring includes one of the following tests at baseline levels and every 2 days.

The test is:

- A. AST and ALT hepatic enzymes
- B. Glomerular filtration rate and creatinine clearance
- C. Complete blood count
- D. Creatinine kinase and troponin
- E. Vitamin B12 levels

iii) What is the first-line drug treatment of this zoonosis?

12. A 12-year-old girl presents to the pediatrician with paroxysmal cough for the 3 weeks. It comes on when she laughs, exercises, or yawns. Her mother says it has a “whooping” sound. Nothing makes it better. It occurs day and night and is interrupting her sleep and her studies. The patient reports that last month “I had a cold with runny nose, mild cough, general unwell feeling, and maybe a slight fever”. An outbreak of *Bordetella pertussis* infection has been reported in the region.

Nasopharyngeal and post-pharynx swabs are obtained for bacterial culture and PCR. The patient is given a clinical diagnosis of whooping cough. An e-prescription is submitted for empiric treatment with a drug that is widely distributed in the body, including lung tissue and macrocytes. The drug is excreted in the feces and has a half-life of ~70 hours, due to sequestration of the drug in tissues. The course of therapy is 5 days with a loading dose on day 1. Due to the long half-life, 5 days of therapy provides a therapeutic effect for up to 10 days.

Which one of the following best describes this drug’s mechanism of action?

- A. Binds 30S subunit → inhibits formation of the initiation complex and causes misreading of the mRNA
- B. Binds 30S subunit → prevents binding of charged aminoacyl tRNA
- C. Binds 50S subunit → prevents the new protein from exiting the ribosome*
- D. Binds 50S subunit → inhibits peptidyl transferase activity and peptide bond formation
- E. Binds 50S subunit → prevents formation of the ribosomal-fMet-tRNA complex

Hint: *B. pertussis* treatment is on p16 of the Notes Handout.

I recommend you go to YouTube and listen to Whooping Cough Sound.

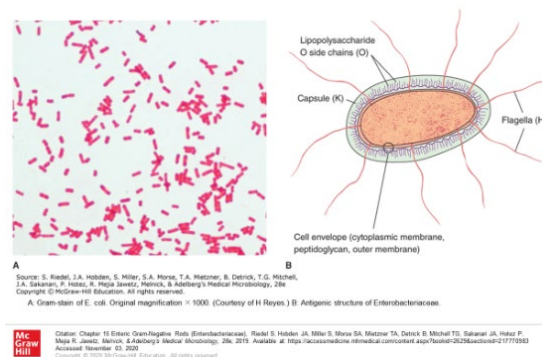
13. Imagine you wish to treat a patient for pertussis (whooping cough) and the supply of the long-acting antibiotic described in the previous question is in shortage due to an environmental disaster at the manufacturing facility.

What is an alternative agent from the same class of antibiotics that could be given?

Short answer

14. A 79-year-old febrile woman with advanced Alzheimer disease, diabetes mellitus type 2 (T2DM or T2D), and recurrent acute urinary tract infections (UTIs) is admitted for treatment of a complicated UTI. She has been receiving cephalexin for UTI prophylaxis for 1 month. Previous UTIs were repeatedly treated with trimethoprim-sulfamethoxazole (TMP-SMX), nitrofurantoin, cephalexin, ciprofloxacin, and ceftriaxone. She has a history of allergy to penicillin (anaphylaxis). The patient resides in the Memory Care unit of a skilled nursing facility.

Symptoms and signs are consistent with urosepsis. Dipstick urinalysis is positive for leukocyte esterase and nitrites, suggestive of infection with Gram-negative bacilli. Urine Gram stain shows Gram-negative rods. Blood and urine cultures are pending.



The patient is placed on empiric intravenous therapy for coverage of *E. coli*, *Klebsiella* spp, and *Proteus mirabilis*. The selected drug is highly effective against most aerobic Gram-negative bacteria. Despite the short 2-hour half-life, once daily IV administration is effective. Because of potential drug toxicity, renal and eighth cranial (vestibulocochlear) nerve function will be closely monitored.

Definitions:

- Complicated UTI refers to a UTI with features suggesting the infection extends beyond the bladder, such as fever, chills or rigors, significant fatigue or malaise, and flank pain.
- Urine nitrite test has a very high sensitivity (>90%). A positive test reflects the presence of Enterobacterales, which convert nitrate normally present in urine to nitrite.

Which one of the following antibiotics fits the activity profile and is likely to be effective in this patient?

- Ampicillin-sulbactam
- Cefotaxime
- Doxycycline
- Gentamicin
- Levofloxacin
- Neomycin
- Sulfamethoxazole

15. A 6-month-old girl is brought to the local emergency department by her mother, who reports that the child has a fever, irritability, sleepiness, doesn't want to eat, and has not vomited. Her family immigrated from Haiti one year ago. Family history reveals that the patient's mother suffered an untoward reaction when she was treated for a urinary tract infection with "a sulfa drug" 6 years ago. The mother was told at the time that she has a G6PD deficiency.

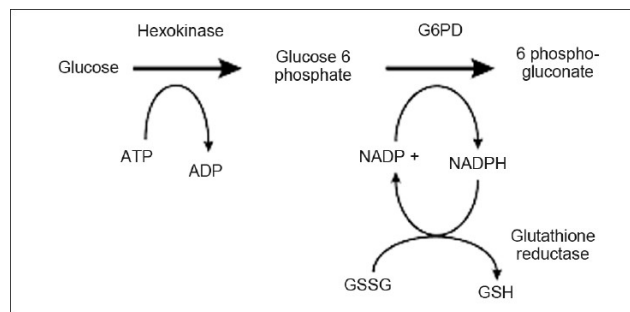
Urinalysis obtained by catheterization is positive for leukocyte esterases and urinary nitrites. Gram stain reveals Gram-negative rods. A urine sample is sent to the microbiology lab for culture.

Based on clinical presentation, the patient is given a diagnosis of uncomplicated urinary tract infection. In view of the family history, trimethoprim-sulfamethoxazole is contraindicated because its potential toxicity in this patient.

What is the adverse drug reaction?

- Hemolytic anemia
- Neutropenia
- Sickle cell anemia
- Thalassemia
- Thrombocytopenia

Definition: G6PD: glucose-6-phosphate dehydrogenase is an enzyme crucial for preventing cellular damage by reactive oxygen species.

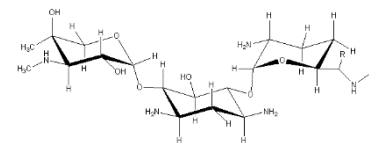


G6PD maintains the body's supply of reduced NADPH. G6PD catalyzes the reduction of NADP+ to the co-enzyme nicotinamide adenine dinucleotide phosphate, NADPH.

16. How do the structure and physicochemical properties of the aminoglycosides contribute to their pharmacokinetic properties:

- parenteral only · distributes in extracellular fluid · does not cross BBB
- renal elimination of unchanged drug by glomerular filtration
- $t_{1/2}$ 2-3 hours, adults, normal renal function · $t_{1/2}$ 24-48 hours, renal impairment

Short answer.



17. An adult patient with a history of dysphagia is admitted for the management of a lung abscess secondary to aspiration pneumonia. The electronic medical record (EMR) indicates an allergy to penicillin (anaphylaxis). The patient is treated with an intravenous antibiotic from one of the following classes because of its effectiveness against the most frequent causative organisms, microaerophilic *Streptococci* species and strict anaerobic species, which are components of the normal flora of the respiratory tract.

Which one?

- Aminoglycosides
- Lincosamides
- Macrolides
- Oxazolidinones
- Streptogramins
- Tetracyclines

Hints:

Think about which antibiotic in this series of lectures is useful against Gram-positive infections and anaerobic infections “above the waist”.

Lung Abscess: A circumscribed area of pus or necrosis in the pulmonary parenchyma, commonly caused by microaerophilic *Streptococci* species and anaerobes *Peptostreptococcus*, *Prevotella*, and *Fusobacterium*. Students, you don’t need to know the names of these anaerobic bugs. “Anaerobic” is the takeaway.

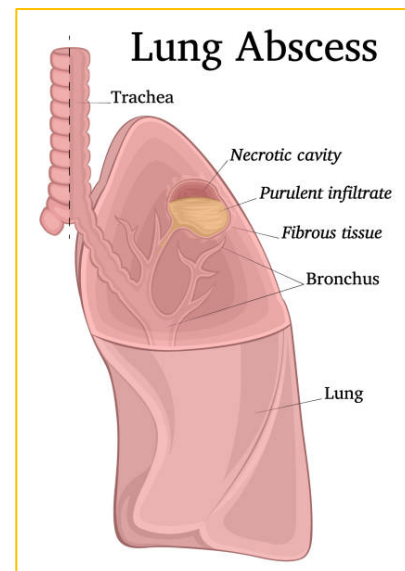


Illustration of a sectioned lung depicting a purulent-necrotic cavity (abscess)
<https://www.istockphoto.com/vector/illustration-of-lung-abscess-gm1124712879-295366028>

18. A 64-year-old man with history of insulin-dependent diabetes mellitus (IDDM) presents to the ED with fever, chills, cough with purulent sputum, and shortness of breath with use of accessory muscles for breathing. He reports allergies (anaphylaxis) to penicillin and ceftriaxone.

- Physical examination, clinical laboratory tests and chest radiograph results are consistent with left lobar pneumonia. Sputum and blood samples are obtained for culture.
- Left lower lobe opacity is noted on chest radiograph.
- The patient is admitted for empiric treatment of pneumococcal pneumonia. He is at low risk for MRSA and *Pseudomonas* infection.

i) Which one of the following options is bactericidal against susceptible strains of the infectious pathogen and is not contraindicated based on this patient's allergy history?

- A. Cefotaxime
- B. Chloramphenicol
- C. Ciprofloxacin
- D. Levofloxacin
- E. Linezolid
- F. Piperacillin-tazobactam

ii) What alternative agent from the same antibiotics class would also be effective?

Short answer

Hint: Think about which class of drugs should be avoided. Then, think about which drugs are effective in treating the likely infective pathogen.



Figure: UpToDate Graphic 6568 version 4.0: Pneumococcal pneumonia chest radiograph.
64-year-old male with insulin-dependent diabetes mellitus. He was admitted with bacteremic pneumococcal pneumonia. Note the left lower lobe opacity.

19. Which one of the following best represents the mechanism of action of the drug selected for the patient in the case vignette above?

The drug:

- A. Causes production of aberrant nonfunctional proteins by misreading mRNA and incorporating incorrect amino acids.
- B. Inhibits the formation of a cofactor that participates in the synthesis of purines and pyrimidines
- C. Introduces DNA strand breaks by preventing the relaxation of DNA supercoils*
- D. Prevents translation of mRNA to form proteins
- E. Triggering autolysin activity and osmotic lysis

20.i) Which one of the following *best* represents the PK-PD profile of the selected drug in the case vignette above?

- A. The antimicrobial concentration remains above the MIC for most of the dosing interval.
- B. A high peak concentration is achieved with a prolonged persistent antimicrobial effect.
- C. The total drug concentration (exposure) over time in relation to the MIC and moderate to prolonged persistent antimicrobial effect.
- D. Mixed PK-PD profile, total drug concentration over time and high peak concentration, are predictors of efficacy.
- E. Mixed PK-PD profile, total drug concentration over time and duration above the MIC are predictors of efficacy.

ii) What information do the pharmacokinetics-pharmacodynamics profiles provide?

21. What are the potential toxicities and the mechanisms associated with fetal adverse effects if the mother is taking a drug from the tetracyclines group for treatment of an infection?

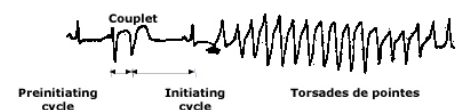
- A. Deformity of the long bones and bone fragility
- B. Hematopoietic disturbances and anemia
- C. Induction of nephrogenic diabetes and polyuria
- D. Kidney stones and obstruction of urine flow
- E. Tooth enamel dysplasia and bone growth disruption

22. A 22-year-old person with a history of schizophrenia treated with ziprasidone (antipsychotic medication) is diagnosed with pneumonia. NKDA. This antipsychotic medication has been associated with QT interval prolongation. Several antimicrobials also increase QT interval. Taking them at the same time increases the risk of ventricular arrhythmia and death.

Which one of the following antimicrobials for empiric treatment of this patient's lung infection is unlikely to increase this risk?

- A. Azithromycin
- B. Ampicillin-sulbactam
- C. Lefamulin
- D. Moxifloxacin
- E. Voriconazole

Single lead electrocardiogram (ECG) showing torsades de pointes



The electrocardiographic rhythm strip shows torsades de pointes, a polymorphic ventricular tachycardia associated with QT prolongation. There is a short, preinitiating RR interval due to a ventricular couplet, which is followed by a long, initiating cycle resulting from the compensatory pause after the couplet.

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23. The local community hospital has seen increasing resistance to doxycycline.

i) PCR analysis shows bacterial expression of tetM gene, which encodes a(an):

- A. Acetylase
- B. Efflux pump
- C. Hydrolase
- D. Methylase
- E. Ribosomal protection protein

ii) What is the potential for cross-resistance due to this resistance mechanism?

Hint: I think you have identified the search term to help you find the answer in the Notes. Try to answer the question before looking up the answer. This practice will help you remember it.

24. A study on fluoroquinolone resistance testing was performed in an ongoing Sexually Transmitted Infections surveillance program in different locations in Kenya between 2013-2017. Antimicrobial Susceptibility Tests were performed in 84 gonococcal isolates; 22 isolates were selected for sequencing. The isolates were subcultured and whole genomes were sequenced. 20 of the isolates were found to be resistant to ciprofloxacin. Gene mutations were identified that conferred reduced affinity of the microbial “receptor” for the fluoroquinolones.

Which of the following proteins was altered due to this gene mutation?

- A. Dihydropteroate synthase
- B. Enzymes that protect the ribosome acceptor site
- C. Lanosterol demethylase
- D. PBP2a
- E. Topoisomerases

25. A 22-year-old woman presents to her PCP with complaints of a nonproductive cough, fever, “bad” headache, muscle aches, fatigue, and photophobia for 3 days. Acetaminophen reduced the fever for 4 to 5 hours but did not relieve the headache or muscle aches. She returned from a visit to her parents’ sheep farm in the South of France 15 days ago. NKDA. Her temperature is 39.1°C (102.3°F). Skin rash is absent. Auscultatory findings are unremarkable. All other physical findings are normal. Suspecting acute Q fever, serologic testing by immunofluorescence assay is performed. Results indicate *Coxiella burnetii* infection.

A drug from what class of antibiotics is preferred for the treatment of this zoonotic infection (Q fever) to shorten the duration of fever and reduce the risk of chronic disease (the goals of therapy).

What class and what is the drug?

- A. Chloramphenicol
- B. Cyclic lipopeptides
- C. Glycopeptides
- D. Lincosamides
- E. Macrolides
- F. Oxazolidinones
- G. Penicillins
- H. Streptogramins
- I. Tetracyclines

Hint: Here is an approach to differentiating the correct answer from the distractors:

Recognize the important clues: Recent visit to sheep farm. A zoonotic infection? Confirmed.

Think about the class of drugs used in the treatment of many zoonoses. You don't need to know the specific bug at this time.

Why is knowing the preferred treatment for zoonotic infections important?

26. A 38-year-old male with extensively drug-resistant (XDR) tuberculosis is given a treatment regimen that includes linezolid. Monitoring includes signs and symptoms of linezolid toxicity that can occur with long-term use of the drug.

What are the potential adverse drug events associated with this drug? (Select all that apply.)

- A. Myelosuppression
- B. Optic neuropathy
- C. Peripheral neuropathy
- D. *C. difficile*-associated diarrhea

Hint: Notes p29.

27. What is the mechanism of action of the drug in the above question (#26)?

One sentence or draw and illustration.

28. What are the spectrum of activity of the oxazolidinone antibiotics and their place in therapy?

29. An adult patient with a complicated intra-abdominal infection is given treatment with intravenous tigecycline targeting Gram-negative coliforms, streptococci, enterococci, and anaerobic bacteria.

i) What is an advantage of this drug related to drug design over other drugs in the same class?

Short answer.

ii) Which of the following is the most likely drug-specific reason some experts avoid the drug?

- A. Concern for increased incidence of mortality associated with use of this drug*
- B. Substantial rates of in vitro resistance to *B. fragilis*
- C. Frequent development of nephrogenic diabetes mellitus
- D. Requirement to avoid the drug in patients with renal impairment.

30. A 44-year-old otherwise healthy person presents to the family physician with s/s of rhinosinusitis of 15 days duration. The symptoms initially improved after one week but reoccured 2 days later. The patient reported an allergic reaction (rash) to Pediazole (erythromycin-sulfisoxazole) as a child and a reaction to amoxicillin 6 months ago (urticaria and angioedema). Based on history and physical findings, an e-prescription for levofloxacin is submitted for presumptive acute bacterial rhinosinusitis.

After taking the drug as directed for 7 days, the patient complains of pain, burning, tingling, numbness, weakness, and a change in sensation to touch and temperature in both hands and both feet.

What drug-related adverse effect is manifested by these signs and symptoms?

- A. Hypersensitivity reaction
- B. Hypoglycemia
- C. Peripheral neuropathy
- D. Rhabdomyolysis

Students: You should know these terms even though this is your first term of medical school. If you do not know what the terms mean, please do a quick Internet search of the definitions.

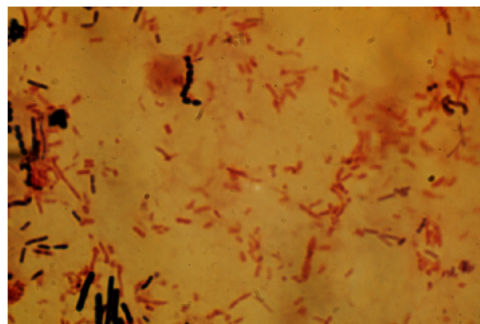
AI can help but be sure the information is accurate before relying on it.

31. A 34-year-old woman with a serious Gram-negative infection is evaluated for empiric therapy that includes an aminoglycoside.

This drug should be avoided (if possible) if the patient has:

- A. Chronic hepatitis C viral infection (liver disease)
- B. Gallstones (potential biliary obstruction)
- C. Myasthenia gravis (neuromuscular disorder)
- D. Warfarin therapy (anticoagulant; high potential for DDIs)

Mixed flora in peritoneal fluid from a ruptured viscus



Gram stain of peritoneal fluid (x1000) shows several different organisms, including gram-positive cocci in chains, gram-positive rods, plump enteric gram-negative bacilli, and thinner gram-negative rods. Mixed fecal flora grew from this specimen.

Courtesy of Harriet Provine.

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32. A 22-year-old person presents to the family physician with a low-grade fever, cough, myalgias, and fatigue. Chest x-ray shows diffuse interstitial infiltrates consistent with atypical pneumonia caused by *Mycoplasma pneumoniae*. The patient is prescribed fluids, rest, ibuprofen or acetaminophen, and an oral antibiotic effective against the bacteria that cause atypical pneumonia and that has a relatively low risk of serious toxicities.

What is the drug?

- A. Amoxicillin
- B. Cefuroxime
- C. Chloramphenicol
- D. Clarithromycin
- E. Linezolid

33. What other classes of drugs would be effective in the treatment of community-acquired pneumonia caused by susceptible atypical bacteria and are generally tolerable?

Make a list.

34. A 45-year-old patient (pronouns he, him, his) presents to the PCP with complaint of occasional nausea and burning epigastric pain 45 minutes after ingesting a meal. He denies the use of aspirin or other NSAIDs, alcohol, and tobacco or vaping products. Past medical history is significant for a particular congenital problem diagnosed when he was 17 years old.

Presenting signs and symptoms are consistent with a peptic ulcer. The urease breath test is positive for *Helicobacter pylori*. Upper endoscopy confirms the diagnosis of benign peptic ulcer disease.

The patient is evaluated for Clarithromycin + Amoxicillin + Lansoprazole (Prevpak®) for eradication of *Helicobacter pylori*. However, the macrolide in the regimen should be avoided due to the patient's congenital abnormality.

Thinking about the potential serious toxicities of the macrolide in the regimen and the patient's hereditary disease, **what is the most likely reason for avoiding this peptic ulcer therapeutic regimen in this patient?**

- A. Agammaglobulinemia (immune deficiency)
- B. G6PD deficiency (metabolic disorder)
- C. Long-QT syndrome (electrophysiological disorder)
- D. Tourette syndrome (neurological disorder)

Hint: Notes p17.

35. Which of the following options best represents the mechanism of action of the aminoglycosides?

- A. Blocking the initiation complex
- B. Eliciting premature protein synthesis termination
- C. Inducing accumulation of nonfunctional monosomes
- D. Misreading mRNA
- E. Incorporating incorrect amino acids
- F. All of the above

What are the aminoglycosides' binding site and its physiologic function?

36. A 72-year-old woman with severe chronic obstructive pulmonary disease (COPD) presents to her pulmonologist for evaluation of frequent exacerbations, despite optimal management. After weighing the potential benefits against the risks of long-term macrolide use, prophylactic macrolide antibiotic therapy is considered for this patient. The patient's medication history is significant for the anticoagulant apixaban for atrial fibrillation. Erythromycin and clarithromycin are contraindicated due to a metabolism and transport related drug-drug interaction with the apixaban.

i) What are the metabolic- and transport-mediated drug interactions that could lead to potentially severe bleeding (an effect of excess systemic levels of the anticoagulant)?

Highlight the two correct answers.

A. CYP1A2	a. OCT
B. CYP2D6	b. GLUT4
C. CYP3A4	c. P-gp
D. UGT1A1	

ii) What is a potentially serious adverse drug event that could occur as a result of this drug-drug interaction?

Hint: Pick out what you *need* to know and what you *do not need* to know.

The information in the stem indicates that there is a potentially serious drug interaction with the apixaban.

To answer this question, you do not need to know the pharmacokinetic properties of the apixaban because you can predict and deduce the potential adverse drug event of the interaction in this case by knowing the potential adverse effects of the apixaban and the metabolism- and transport-related interactions caused by the macrolide antibiotic.

Of course, in practice, you will always look it up in the drug monographs to be sure.

37. After checking the Drug Interactions Tool in UpToDate, an e-prescription is submitted for a macrolide that does not inhibit CYPs or inhibit renal transporters.

i) What antibiotic is selected for the COPD patient in the above question? Short answer

ii) What are the clinically relevant pharmacokinetic properties of this drug?

38. Janie is 20-year-old person who presents to the PCP with complaint of a “puss-filled sore” on their left elbow. It began as an abrasion that has gradually gotten worse over the past week. The patient has not had a fever or any other systemic symptoms.

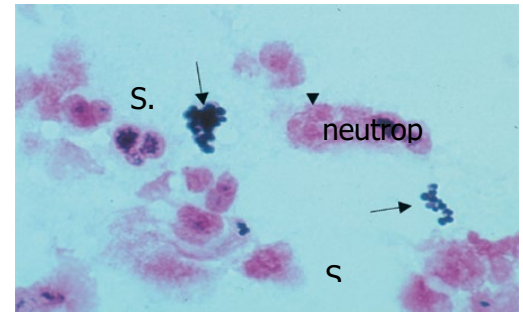
The abscess is debrided and a sample of exudate is sent for culture.

(Debride pronunciation: de breed’)

Gram stain shows Gram-positive cocci in clusters resembling bunches of grapes (arrows) and neutrophils (arrowhead).

An e-prescription is submitted for empiric therapy with an oral protein synthesis inhibitor. The drug:

- Accumulates in phagocytes and wound exudates
- Inhibits peptidyl transferase activity, which prevents bond formation with the incoming amino acid,
- Has good activity against community-acquired MRSA (caMRSA).



Source: Kenneth J. Ryan: Sherris Medical Microbiology, Seventh Edition Copyright © McGraw-Hill Education. All rights reserved.

What is the drug?

- A. Ciprofloxacin
- B. Clindamycin
- C. Erythromycin
- D. Sulfadiazine
- E. Tetracycline

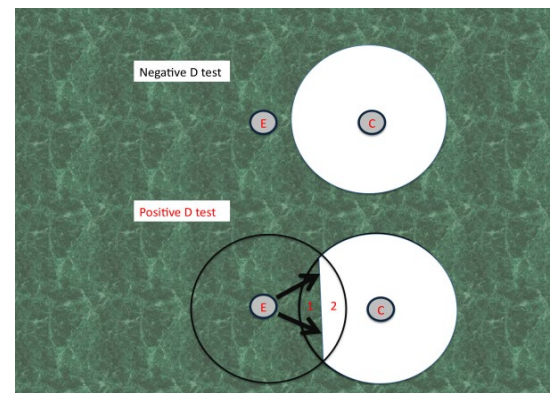
39. Janie initially responds to antibiotic therapy. D-test result obtained the following day is positive, which suggests that a subpopulation of MRSA resistant to Janie’s prescribed antibiotic may lead to clinical failure or recurrence of the infection after therapy is discontinued.

i) What are the gene family and the expressed protein?

ii) What bacteria may harbor these genes?

iii) What is the mechanism of resistance?

iv) What three classes of drugs are cross resistant?



40. A 28-year-old patient in the burn unit is treated for severe thermal burn injury and sepsis. NKDA. Management includes empiric broad-spectrum antibiotic therapy and debridement of the necrotic tissue. Based on the hospital’s antibiogram, a 3-drug regimen is selected for coverage of presumptive MRSA and other Gram-positive bacteria, enteric aerobic Gram-negative bacilli (GNBs), and multidrug-resistant *P. aeruginosa*.

What antibiotic therapy is being given to this patient?

Highlight the preferred agent from each column in the table below (one choice each).

A. Aerobic Gram-negative bacilli and <i>P. aeruginosa</i>	B. Broad-spectrum agent with activity against aerobic GNBs and <i>P. aeruginosa</i>	C. Anti-MRSA
Amikacin Gentamicin Neomycin Paromomycin Tobramycin	Ampicillin-sulbactam Ceftriaxone Clindamycin Meropenem Tigecycline	Cefazolin Daptomycin Linezolid Quinupristin-Dalfopristin Vancomycin

41. A 26-year-old patient presents to the ED with fever, cough, and shortness of breath. Chest radiograph shows diffuse bilaterally symmetric interstitial pattern opacification and multiple small hyperlucent patches. HIV screening is positive. The patient is referred to the infectious disease specialist and ultimately given a diagnosis of HIV/AIDS (CD4 cell count 150 cells/mm³) and *Pneumocystis* pneumonia, a fungal infection.

Antiretroviral therapy and the drug of choice for *Pneumocystis jirovecii* pneumonia treatment (and prophylaxis) are initiated.

The patient presents 7 days later with a rash that appeared suddenly this morning. Suspecting an exanthematous drug eruption, the patient is directed to stop taking the drug for the pneumonia treatment. The rash resolves gradually over the course of the next 5 days. He is placed on an alternative antimicrobial regimen.

What drug is the preferred treatment for *Pneumocystis* pneumonia that likely caused the rash?

- A. Amphotericin B
- B. Ciprofloxacin
- C. Doxycycline
- D. Itraconazole
- E. Trimethoprim-Sulfamethoxazole

Hint: Handout p5-6



Diffuse bilaterally symmetric interstitial pattern is noted in the perihilar region and extending towards the periphery. Multiple ill-defined small hyperlucent patches are noted in the bilateral lung fields especially in the mid-zones suggestive of pneumatocele.



Numerous small, discrete papules on the back of a patient with exanthematous drug eruption.

42. In addition to skin rashes, the hypersensitivity reaction to the sulfonamide may progress. What other manifestations may occur? (More than one answer. Choose all that apply.)

- A. Kidney: Crystalluria
- B. Liver: Hepatitis
- C. CNS: Kernicterus
- D. Skin: Stevens-Johnson syndrome
- E. Bone marrow: Thrombocytopenia

Definition: Morbilliform (looks like measles) exanthem (rash); macular lesions.

43. What is an important metabolic enzyme primarily involved in the hepatic biotransformation of sulfonamides to inactive metabolite that is subject to genetic polymorphisms resulting in slow, intermediate, and rapid metabolizer phenotypes?

Short answer

44. A penicillin- and cephalosporin-allergic health care worker is treated for a subcutaneous *S. aureus* infection with clindamycin 450 mg po TID x10d. The infection resolves and the patient completes the clindamycin therapy. Two weeks later, the patient calls his physician with complaint of low-grade fever, abdominal cramping, loss of appetite, and diarrhea consisting of 6-8 watery stools per day for 3 days. He denies blood in the stool. Stool cultures reveal an organism known to cause antibiotic-associated diarrhea. Nucleic acid amplification test reveals bacterial toxin.

Definition: po TID: by mouth 3 times daily

What is the diarrhea-causing pathogen in this case?

- A. *Clostridioides difficile*
- B. *Bacteroides fragilis*
- C. *Enterococcus faecium*
- D. *Escherichia coli*
- E. *Salmonella enterica*

45. A generally healthy 42-year-old patient, who has been given a clinical diagnosis of acute cystitis, is evaluated for antibiotic therapy effective against *E. coli*. The patient has never before had a urinary tract infection. Medical history is significant for warfarin prophylaxis of venous thromboembolism following knee arthroplasty two weeks ago. The patient does not smoke or use nicotine products, drink alcoholic beverages, or use recreational drugs. NKDA. Trimethoprim-sulfamethoxazole should be avoided in this patient.

What is the reason for selecting an alternative antibiotic instead of TMP-SMX?

- A. TMP-SMX could be ineffective in treating urinary tract infections caused by *E. coli*.
- B. TMP-SMX could crystallize in renal tubules and cause obstruction.
- C. TMP-SMX could increase the risk of excessive bleeding.
- D. TMP-SMX could cause emergence of *S. aureus* resistant strains and cause surgical site infection.

Hint: Notes p11

46. What are the postulated mechanisms of the TMP-SMX and warfarin drug-drug interaction in the above scenario?

State 3 possible mechanisms may occur simultaneously.

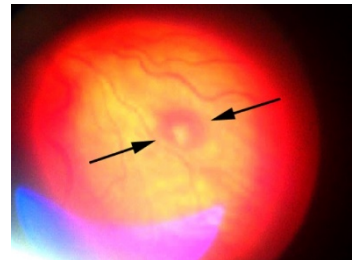
Hint: Handout p11

47. A 45-year-old woman presented to the ED with fever, shortness of breath, persistent back pain, and a 10-pound weight loss. She reported chills, sweats, and anorexia for 6 weeks, which had been progressively increasing in severity, and persistent back pain for 4 weeks. Past medical history was significant for rheumatic fever, characterized by swollen joints and fever, which confined her to bed for 3 months when she was a young child. NKDA.

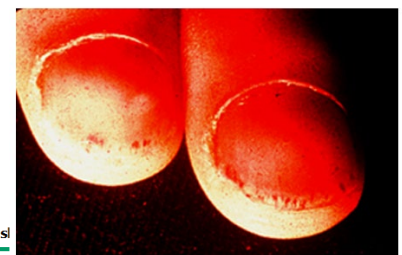
- VS: T 100.4°F (38°C), pulse 90/min, resp 18/min, BP 130/80 mmHg;
- Roth spot (a round white spot surrounded by hemorrhage) in the retina of her right eye and petechiae were seen in the conjunctiva of both eyes.
- Splinter hemorrhages were seen under two fingernails of her right hand and one finger of the left hand.
- Osler nodes (tender, small, raised, red or purple lesions of the skin) were seen in the pads of one finger and one toe.
- Her heart size was normal to percussion.
- On auscultation, a low-pitched diastolic murmur consistent with mitral valve stenosis was heard at the apex; a loud mitral valve opening snap was heard over the left chest.

https://www.google.com/search?q=mitral+valve+stenosis+spund&source=lmns&tbn=vid&bih=783&biw=1600&rlz=1C1CHBF_enUS1029US1029&hl=en&sa=X&ved=2ahUKEwiqw_av0rmAAxWzUDUKHa63CpAQ0pQJKAF6BAGBEAQ#fpstate=ive&vld=cid:ae221fbf,vid:5oCptZo4pUY

- Echocardiography showed an enlarged left atrium, thickened mitral valve leaflets, and a vegetation on the posterior leaflet.



Subungual splinter hemorrhages



Subungual splinter hemorrhages in a patient with *Trichinella papuae*.

Courtesy of Peter F Weller, MD.

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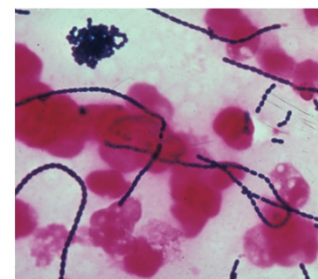
Osler nodes are tender papulopustules located on the pulp of the finger in a patient with bacterial endocarditis caused by *Staphylococcus aureus*.

Courtesy of Charles V Sanders. *The Skin and Infection: A Color Atlas and Text*, Sanders CV, Nesbitt LT Jr (Eds), Williams & Wilkins, Baltimore 1995.

<http://www.lww.com>

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- 3 blood samples were obtained for culture. The patient was admitted for treatment.
- One day later, all three were positive for Gram-positive cocci in chains that were viridans group streptococci subsequently identified as *Streptococcus sanguis*.
- Sensitivity results show that the organism was sensitive to the antibiotics tested but the MIC for the beta-lactam was elevated indicating intermediate resistance.



Source: S. Riedel, J.A. Hobbs, S. Miller, S.A. Morse, T.A. Nietzner, B. Dietrich, T.G. Mitchell, J.A. Sakamoto, P. Hostet, R. Meja Jawetz, Mallick, & Adenborg's Medical Microbiology, 28e Copyright © McGraw-Hill Education. All rights reserved.

Streptococci grown in blood culture showing Gram-positive cocci in chains. Original magnification $\times 1000$.

Antibiotic treatment: 2-drug intravenous antibiotic therapy was initiated.

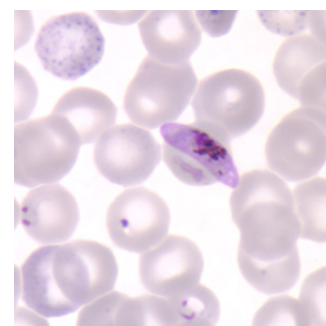
- One drug was a cell wall inhibitor with a relatively narrow spectrum that is highly active against viridans group streptococci – in fact, highly active against all susceptible streptococci.
- The second drug was able to reach its intracellular site of action because of the breakdown of the cell wall and was added for synergistic effect.

What antibiotics were administered? Highlight the selected agents:

Cell wall inhibitor	Drug for synergy
Aztreonam	Azithromycin
Cefiderocol	Doxycycline
Imipenem-cilastatin	Gentamicin
Ampicillin	Vancomycin

Hint: Do not be distracted by the name of the specific bacterial pathogen. Focus on the relevant information:

48. A 6-year-old child is brought to the ED with complaints of fever, chills, muscle aches, night sweats, and malaise. His family recently returned to the US from a malaria endemic region. Giemsa-stained blood smears by light microscopy show *Plasmodium falciparum* (malaria). He is given a 7-day course of therapy with quinine and an antibiotic effective in the treatment of malaria.



What is the prescribed drug?

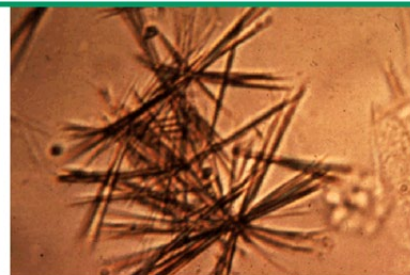
- Chloramphenicol
- Clarithromycin
- Clindamycin
- Doxycycline
- Linezolid

49. A 78-year-old woman presents with severe flank pain, tachycardia, diaphoresis, and nausea. She takes a urinary antiseptic that is hydrolyzed to formaldehyde for prophylaxis of recurrent urinary tract infections. She explained that she is also self-treating acute urinary symptoms that started 3 days ago with TMP-SMX, which she had left over in her medicine cabinet. Physical examination and urinalysis findings are consistent with crystalluria.

i) **Which one of the following is the urinary antiseptic?**

- A. Ampicillin
- B. Cephalexin
- C. Ciprofloxacin
- D. Methenamine
- E. Sulfasalazine

Photomicrograph showing urine sediment of a patient with sulfonamide crystalluria



Urine sediment showing sulfonamide crystals with a needle-shaped appearance. Other forms that may be seen include rosettes and a shock of wheat appearance.

Courtesy of Harvard Medical School.

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ii) **What is the mechanism of this drug-induced adverse event?**

50. A 50-year-old man with amyotrophic lateral sclerosis is admitted to the ICU for the management of respiratory distress. He has been a resident of a skilled nursing facility for 2 years. NKDA. The patient is placed on a mechanical ventilator.

On Hospital Day 5 he develops a fever, leukocytosis, increase in respiratory secretions, pulmonary consolidation on PE, and radiographic infiltrates. The patient is evaluated for antibiotic therapy.

Potential infective organisms are MDR bacteria: Gram-negative bacilli, *P. aeruginosa*, pneumococcus, and *S. aureus*. The hospital's antibiogram indicates that MDR *P. aeruginosa* is prevalent in the hospital.

Definition: Antibiogram: a report that shows the susceptibility of pathogenic strains to antibiotics

i) **What are the potential mechanisms of resistance? Highlight all that apply.**

ii) **Which resistance mechanism is most common?**

Deletion of porins
Conditions of low oxygen tension
Expression of MLS _B methylases
Microbial transferases phosphorylate, adenylate, or acetylate the drug
Ribosomal proteins with low affinity for the drug
Thickening of the peptidoglycan wall

51. i) What are the three major toxicities associated of aminoglycosides? Short answer

ii) What are the mechanisms of these adverse effects? Short answer

iii) What tests can be used to monitor for these toxicities? Short answer

52. Gentamicin- and tobramycin-resistant Gram-negative bacteria are usually susceptible to which alternative drug in the same class? Why?

Short answer

53. A hospitalized patient is given intravenous linezolid for the treatment of an infection caused by multidrug resistant:

- A. *Bacteroides fragilis*
- B. *Clostridium difficile*
- C. *Escherichia coli*
- D. *Streptococcus pneumoniae*
- E. *Pseudomonas aeruginosa*

Clue:

The student who recognizes the specific Gram-positive and Gram-negative bacteria will easily come to the correct answer, as only one Gram-positive organism is listed.

54. The patient in the previous question is concurrently taking an antidepressant that, together with linezolid, can cause a potentially serious adverse drug event manifested by confusion, tachyarrhythmia, hyperthermia, and seizure. The antidepressant increases levels of 5-hydroxytryptamine in the neuronal synapses in the brain. The antibiotic inhibits a specific enzyme that inactivates this monoamine neurotransmitter, further increasing synaptic levels.

The patient will be monitored for signs and symptoms of:

- A. Acute kidney injury
- B. Cardiotoxicity
- C. Hemolytic anemia
- D. Hepatitis
- E. Serotonin syndrome
- F. Skeletal muscle toxicity

Hint: Handout p29-30.

Hint: 5-hydroxytryptamine has another name. Look it up.

55. An 83-year-old woman tells her gerontologist about a time many years ago when she had to go to the emergency room because of fatigue, weakness, and mental confusion. They told her she had, “some kind of kidney acidosis”. She was told that it apparently resulted from a drug reaction caused by taking leftover antibiotic “to cure my cold. It must have been in my medicine cabinet for 6 or 7 years. I saved it in case I needed it,” she reported.

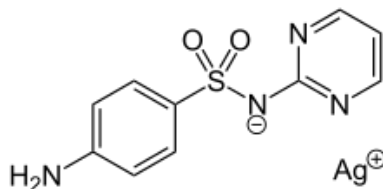
Which of the following drugs is associated with this toxicity?

- A. Ampicillin
- B. Cephalexin
- C. Chloramphenicol
- D. Erythromycin
- E. Tetracycline

56. A 12-year-old male is admitted for the treatment of second- and third-degree burns on his left arm. His therapy includes an antimicrobial dressing to prevent wound sepsis. The selected sulfonamide is associated with silver, which is released slowly and is selectively toxic to microbes.

What is the sulfonamide component in the dressing?

- A. Mafenide
- B. Sulfacetamide
- C. Sulfadiazine
- D. Sulfadoxine
- E. Sulfamethoxazole
- F. Trimethoprim



57. MATCHING Differentiating the clinical applications of the aminoglycosides.

Match the drugs with their places in therapy:

Activity against organisms harboring plasmid-mediated transferases	Streptomycin
Combination therapy	Gentamicin
Complicated UTI with no alternative treatment options	Tobramycin
Gram-negative bacilli	Amikacin
Infective endocarditis	Plazomicin
Inhaled for lung infection in cystic fibrosis patients	Neomycin
Intestinal <i>Entamoeba histolytica</i> infection	Paromomycin
Not used for systemic infections	All AGs
<i>Pseudomonas aeruginosa</i>	
Reduction of aerobic bowel flora	
Tuberculosis	
Tularemia	

58. **Review:** In the treatment of outpatients without comorbidities, what classes of protein synthesis inhibitors would also be effective against susceptible strains of the potential causes of community-acquired pneumonia: “atypicals” and pneumococcus?

Short answer

59. **Review:** For outpatients whose community-acquired pneumococcal pneumonia can be treated with an oral agent and who are not allergic, what oral drug in the penicillins class would be useful?

Short answer

Learning tip: SPACED RETRIEVAL PRACTICE

Studying information more than once and leaving considerable time between practice sessions promotes durable learning. The example here is retrieval practice on beta-lactam antibiotics.